SINGAPORE DIMENSIONS • GRADE 7A • CHAPTER 4

Linear Equations & Stage Balancing

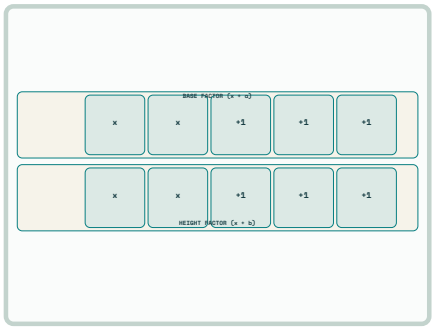
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.7.EE.B.4

Student Name: _____ Date: _____ **Model: SC-32** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-32 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Algebra Stage

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place 3x tiles + 4 unit squares on left rail; place 19 unit squares on right rail.

Sketch Model on Paper:
Draw two-pan balance scale representing 3x + 4 = 19.

3. Step-by-Step LCD Stack Trace (SC-32)

MODE: 123 Base-10 Arithmetic

1.	3	Step 1	Execute [3]
2.	1	Step 2	Execute [1]
3.	enter	Step 3	Execute [enter]
4.	7	Step 4	Execute [7]

EXPECTED LCD X: x = 6

1 **Solve 5x + 12 = 47.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

2 **Solve 3x - 8 = 25.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

3 **Solve 6x + 15 = 63.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

4 **Solve 2x - 14 = 28.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

Station Diagnostic & Solution Key: Chapter 4

Linear Equations & Stage Balancing • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Algebra Studio & Factoring Stage

CAD Blueprint: algebra_stage

Target Device: SC-32 Scientific

CAD / STL Source File:

Education/scad/algebra_factoring_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal Interlocking L-Rails & White Tiles (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Solve 5x + 12 = 47.** Standard: CCSS.MATH.7.EE.B.4	x = 7	[47] [ENTER] [12] [-] [5] [÷]
2	**Solve 3x - 8 = 25.** Standard: CCSS.MATH.7.EE.B.4	x = 11	[25] [ENTER] [8] [+] [3] [÷]
3	**Solve 6x + 15 = 63.** Standard: CCSS.MATH.7.EE.B.4	x = 8	[63] [ENTER] [15] [-] [6] [÷]
4	**Solve 2x - 14 = 28.** Standard: CCSS.MATH.7.EE.B.4	x = 21	[28] [ENTER] [14] [+] [2] [÷]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Remind students that whatever operation is performed on one side must be performed on the other.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-32, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.